I understand. I am acting as a Principal Platform Architect designing the **Self-Hosted Platform Update & Regulatory Distribution Strategy**. This strategy will dictate how the licensed DTE software remains compliant across hundreds of isolated, customer-controlled deployments over a decade of operation, independent of specific implementation technologies.
Here is the definitive strategy.

Self-Hosted Platform Update & Regulatory Distribution Strategy

1. Update Philosophy

The foundational philosophy of this platform is the **Strict Separation of Software from Regulation**.
To survive a decade of regulatory drift across isolated, customer-hosted environments, the system must treat regulatory changes as *data updates*, not *code updates*. If an MH catalog changes, the software should not require a new compiled version.

1.1 What Must Be Updateable

- **Regulatory Data (Dynamic):** MH Catalogs, schema definitions, validation rule sets, DTE type specifications, and retention policies. These must be updatable asynchronously from the core software.
- **Software Execution (Static):** The core issuance engine, evidence preservation mechanisms, audit trail logic, and cryptographic signing routines. These are updated via traditional software releases.

1.2 What Must Never Be Updateable

- **Historical Truth:** Immutable artifacts (Evidence Bundles), the Audit Trail, and previously assigned Control Numbers. Updates must never retroactively alter or invalidate historical documents or their associated regulatory context at the time of issuance.
- **Issuer Identity:** The core cryptographic capability and tax identity of the customer.

1.3 The Core Principle

Software updates deliver capabilities; Regulatory updates deliver compliance. A customer running software v1.2 should be able to receive regulatory bundle v2026.05 and issue compliant documents without upgrading their software version, provided the software capabilities satisfy the schema requirements.

2. Regulatory Distribution Model

The vendor produces **Regulatory Bundles**—digitally signed, self-contained packages containing the current state of El Salvador's DTE compliance rules.

2.1 The Regulatory Bundle

A Regulatory Bundle is a comprehensive, versioned payload containing:

- **Catalog Packages:** The complete append-only set of MH Catalogs, including newly effective entries and deprecated historical entries.
- **Schema Packages:** The structural definitions (fields, types, required flags) for all active DTE Types (e.g., 01, 03).
- **Validation Packages:** Declarative rule sets governing cross-field constraints, totals reconciliation, and catalog membership enforcement.
- **Regulatory Metadata:** Defined legal windows (e.g., maximum contingency duration) and effective dates.

2.2 Distribution Mechanics

The vendor maintains a central distribution authority. Customer deployments connect to this authority to pull bundles.

- **Integrity:** Every bundle must be cryptographically signed by the vendor. The customer platform must reject any bundle lacking a valid signature to prevent tampering.
- **Idempotency:** Applying the same bundle twice must have zero side effects.
- **Delta vs. Full:** Bundles should be cumulative. A customer deployment that has been offline for six months must be able to pull the latest bundle and achieve full compliance without applying intermediate updates sequentially.

3. Update Channels

Deployments exist in wildly varied customer environments. The update strategy must accommodate different levels of access and control.

3.1 Automatic Updates (The Default)

The platform periodically polls the vendor's distribution authority. When a new Regulatory Bundle or minor software patch is available, it is downloaded, verified, and applied automatically during a defined maintenance window.

3.2 Semi-Automatic Updates (Regulated Environments)

For customers requiring change control (e.g., banks, large enterprises), the platform detects updates but requires an explicit approval action by an Owner role before application.

3.3 Manual / Offline Updates (Air-Gapped)

For environments without outbound internet access to the vendor, the platform must support manual ingestion. An authorized user downloads a signed Regulatory Bundle or Software Package from a vendor portal and uploads it directly into the platform via the UI.

4. Versioning Strategy

A robust versioning strategy prevents catastrophic mismatches between software capabilities and regulatory demands.

4.1 Platform Versioning (SemVer)

The software versioning follows semantic rules (Major.Minor.Patch).

- **Major:** Structural changes (e.g., V1 to V2). May require data migrations.
- **Minor:** New capabilities (e.g., adding Email Delivery).
- **Patch:** Bug fixes.

4.2 Regulatory Versioning (Date-Based)

Regulatory Bundles are versioned chronologically (e.g., YYYY.MM.DD.Revision). This provides immediate clarity on compliance freshness.

4.3 Compatibility Guarantees (The Matrix)

Every Regulatory Bundle must declare a MinimumPlatformVersion.

- *Scenario:* MH introduces a minor catalog addition. The new Regulatory Bundle declares MinVersion: V1.0.0. All V1 customers can consume it.
- *Scenario:* MH introduces a radically new cryptographic requirement. The new Regulatory Bundle declares MinVersion: V1.5.0. An older platform will reject the bundle and alert the user that a software upgrade is mandatory for continued compliance.

5. Backward Compatibility Model

The platform must maintain the ability to read, render, and prove the legal validity of a document years after the schemas and catalogs have changed.

5.1 Append-Only Regulatory Data

As defined in the Catalog Model, regulatory updates are append-only.

- **Old Documents:** A document signed in 2024 against Schema v1 and Catalog Cat-011 remains perfectly intact. The system uses the document's embedded SchemaVersion and issuance timestamp to resolve the correct validation rules and catalog meanings for that specific point in time.
- **Historical Validation:** The system must always be able to validate a document against the rules that were active *when it was issued*, not the rules active today.

6. Deployment Lifecycle

The lifecycle manages the safe progression of software and regulatory state.

6.1 Fresh Installation

The installer lays down the core software. The first mandatory operational step is acquiring the baseline Regulatory Bundle (via network or manual upload). Issuance is locked until a valid bundle is applied.

6.2 Upgrade

Upgrades are transactional.

1. **Software Upgrade:** Binaries/services are updated. Data migrations execute. If successful, the new version activates.
2. **Regulatory Upgrade:** The bundle is parsed, signatures verified, and new configurations are staged. The EffectivePeriod data model dictates when the new rules activate.

6.3 Rollback & Recovery

- **Software Rollback:** Supported via standard deployment practices (e.g., reverting binaries and restoring the database backup).
- **Regulatory Rollback:** Generally prohibited. If MH mandates a rule, reverting it risks issuing non-compliant documents. However, if a vendor-issued bundle contains an error, a new bundle with a newer chronological version and corrected rules is pushed. The platform always honors the highest valid version.

7. Failure Scenarios

The system must fail safely, preserving regulatory integrity over operational uptime.

- **Failed/Partial Update:** Regulatory bundles must be applied atomically. If parsing or validation fails midway, the transaction aborts, and the system reverts to the previous valid bundle. An Audit Entry logs the failure.
- **Corrupt/Tampered Package:** The cryptographic signature verification fails. The package is discarded, and a critical alert is generated.
- **Unsupported Schema/Rule:** The customer attempts to load a Regulatory Bundle requiring Platform V2 onto a V1 installation. The system rejects the bundle cleanly, noting the version mismatch.
- **Disconnected Customer:** A deployment cannot reach the vendor for 90 days. The platform continues to operate using its last known valid bundle but surfaces prominent warnings to the Owner regarding potential compliance drift.
- **Outdated Customer:** The platform detects that a newer Regulatory Bundle is available but has not been applied (either due to network failure or manual approval delays). If the system attempts to issue a document using rules that are now officially deprecated by a newly effective mandate, the issuance must be blocked.

8. Compliance Assurance Strategy

The platform must actively monitor its own compliance posture.

- **Drift Detection:** The system continuously compares its active Regulatory Bundle version against the latest version broadcast by the vendor.
- **Pre-Flight Checks:** Before any document transitions to Validated, the system asserts that the selected schemas, catalogs, and rules are Active based on the current timestamp.
- **Unsupported DTE Prevention:** If an issuer attempts to compose a DTE type (e.g., FEX 11) that is not defined in the active Regulatory Bundle (because they are on V1), the Type Registry rejects the request immediately.
- **Alerting:** Imminent regulatory deadlines (e.g., "MH Schema V1 expires in 30 days") trigger progressive alerts in the UI, requiring Owner acknowledgment.

9. Multi-Year Sustainability

To survive a decade, the architecture assumes continuous, compounding change.

1. **Never Hardcode Policy:** No tax rates, schema fields, or catalog codes are ever hardcoded in the software execution paths.

2. **Data Isolation:** The physical isolation of evidence (the EvidenceBundle) guarantees that no future software update or regulatory bundle can retroactively corrupt the proof of a historical transaction.
3. **The Registry Seam:** All Issuance logic relies on the Document Type Registry. When MH introduces Schema v4 in 2030, the vendor publishes a new Regulatory Bundle containing the v4 specification. The platform ingests it, validates it, and the Issuance engine simply points to the new specification. The core software remains untouched.

By adhering to this strategy, a vendor can safely license this software, distribute updates globally, and guarantee that thousands of isolated deployments remain compliant with El Salvador's Ministry of Finance, regardless of how drastically the regulations evolve over the next ten years.